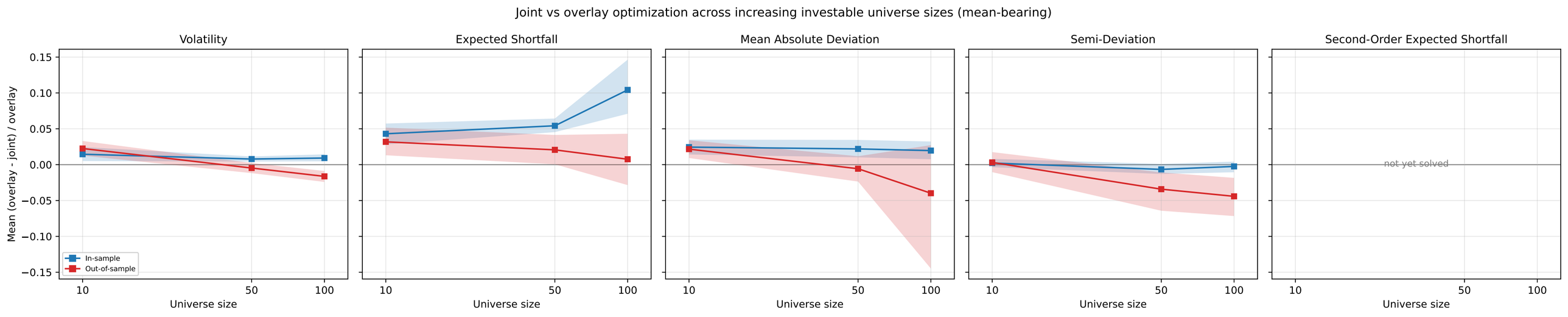
Joint vs overlay optimization across increasing investable universe sizes (mean-bearing)
Volatility
Expected Shortfall
Mean Absolute Deviation
Semi-Deviation
Second-Order Expected Shortfall
Mean (overlay - joint) / overlay
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
In-sample
Out-of-sample
not yet solved
10
50
100
Universe size